



Department of Information Technology

Academic Year 2024 - 2025 (Even Semester)

Degree, Semester & Branch: IV Semester B.Tech. IT

Course Code & Title: CS3452 & Theory of Computation

Name of the Faculty member (s): Dr.V. Anusuya, ASCP/IT

Innovative Practice Description

- **Unit / Topic:** Unit I / Conversion of NFA into DFA, ϵ -NFA to NFA, ϵ -NFA to DFA
- **Course Outcome:** CO1
- **Topic Learning Outcome:** TLO2, TLO3
- **Activity Chosen:** Think Pair Share
- **Justification:**

In unit I, Conversion of NFA into DFA, ϵ -NFA to NFA, ϵ -NFA to DFA is an important topic in the unit. This activity helps the students to know the concepts and conversion from ϵ -NFA to NFA, ϵ -NFA to DFA.

- **Time Allotted for the Activity:** 10 Minutes
- **Details of the Implementation:**
 - The topic NFA to DFA taught, ϵ -NFA to NFA, ϵ -NFA to DFA and NFA to DFA in the previous classes. The time allotted for the activity is 10 Minutes.
 - A team of three members assigned one problem statement in any one of conversion i.e ϵ -NFA to NFA, ϵ -NFA to DFA and NFA to DFA as shown in Figure 1 ,2 and 3
 - The team members discussed and completed the conversion as shown in Figure 4.
 - The students Divya Sri J S presented ϵ -NFA to NFA conversion and Parasuram CA presented ϵ -NFA to DFA as shown in Figure 5.

• **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO8	PO9	PO10	PO12	PSO1	PSO2	PSO3
CO1	3	3	3	1	2	1	2	1	2	2	2

(1 – Low 2 – Moderate 3 – High)

• **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO3	PO4	PO8	PO9	PO10	PO12	PSO1	PSO2	PSO3
	3	3	3	1	2	1	2	1	2	2	2
Justification for correlation	Apply basic Knowledge and fundamentals of Theory of Computation for converting NFA to DFA.	The students will be able to identify the need for analysis of complex engineering problems	Able to design the process flow for converting NFA to DFA	Able to analyze the data for ϵ -NFA to NFA, NFA to DFA, conversion	The students will be able to follow ethics for solving problems	The students able to function as a team for converting ϵ -NFA to NFA, NFA to DFA	The students will be able to communicate one another and can able to present	The students able to learn and implement the concept in Natural Language Processing	The students able to apply mathematical and computing skills.	The students can exhibit knowledge of languages in the field of Cloud Computing to develop IT solutions	The students able to exhibit knowledge in Languages to develop a quality product.

- Images / Screenshot of the practice:



Figure 1: Think Pair Share Activity

NFA to DFA

1. Construct the DFA equivalent to the NFA
 $M = (\{p, q, r\}, \{0, 1\}, \delta, p, \{q, s\})$
 Where δ defined in the following table.

Δ	0	1
P	$\{q, s\}$	$\{q\}$
Q	$\{r\}$	$\{q, r\}$
R	$\{s\}$	$\{p\}$
S	-	$\{p\}$

2. Construct a DFA equivalent to the NFA given below.

Δ	0	1
P	$\{p, q\}$	p
Q	r	r
R	s	-
S	s	s

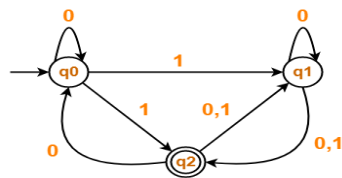
3. Convert the following ϵ NFA to a DFA using the subset construction algorithm.

Δ	ϵ	a	b	c
$\rightarrow p$	ϕ	$\{p\}$	$\{q\}$	$\{r\}$
Q	$\{p\}$	$\{q\}$	$\{r\}$	ϕ
$*r$	$\{q\}$	$\{r\}$	ϕ	$\{p\}$

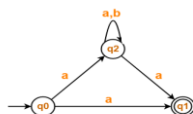
4. Convert the following NFA to its equivalent DFA



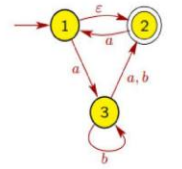
5. Convert the following NFA to DFA



6. Convert the following NFA to DFA



7. Convert it into DFA



8. Convert the following NFA to DFA

	0	1
$\rightarrow p$	$\{q, s\}$	$\{q\}$
$*q$	$\{r\}$	$\{q, r\}$
r	$\{s\}$	$\{p\}$
$*s$	ϕ	$\{p\}$

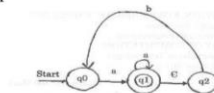
9. Convert the given NFA to DFA

	0	1
$\rightarrow p$	$\{p, q\}$	$\{p\}$
q	$\{r, s\}$	$\{t\}$
r	$\{p, r\}$	$\{t\}$
$*s$	ϕ	ϕ
$*t$	ϕ	ϕ

10. Construct DFA equivalent to NFA $\{\{p, q, r, s\}, \{0, 1\}, \delta, p, \{s\}\}$ where δ is defined as

	0	1
$\rightarrow p$	$\{p, q\}$	$\{p\}$
q	$\{r\}$	$\{r\}$
r	$\{s\}$	-
s	$\{s\}$	$\{s\}$

11. Convert the ϵ -NFA to DFA



12. Determine DFA from the given NFA.

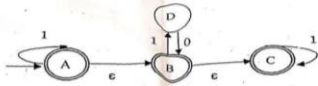
$M = (\{q_0, q_1\}, \{0, 1\}, \delta, q_0, \{q_1\})$ where δ is given by
 $\delta(q_0, 0) = \{q_0, q_1\}, \delta(q_0, 1) = \{q_1\}, \delta(q_1, 0) = \phi, \delta(q_1, 1) = \{q_0, q_1\}$

13. Construct an NFA without ϵ -transitions for the NFA give below.



14. Convert the following ϵ -NFA to NFA and then convert the resultant NFA to DFA.

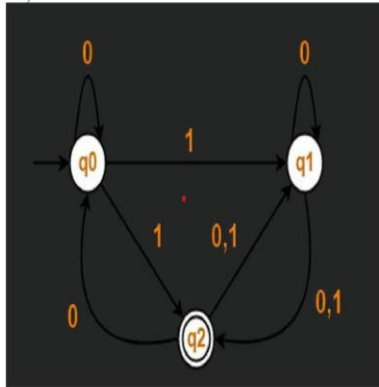
Figure 2 Problems given to Students



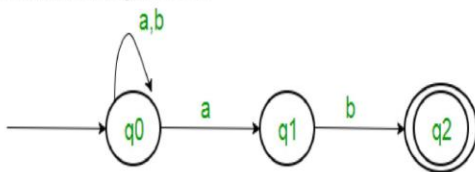
15. Consider NFA $N=(Q,\Sigma,\delta,Q,F)$ where $Q=\{1,2,3\}$, $\Sigma=\{a,b\}$, $q=1$, $F=\{2\}$ and δ is given by the following table.

	a	b	c
1	3	$\varnothing$	2
2	$\{1\}$	$\varnothing$	$\varnothing$
3	$\{2\}$	$\{2,3\}$	$\varnothing$

16. Convert the following Non-Deterministic Finite Automata (NFA) to Deterministic Finite Automata (DFA)



17. Consider the following NFA to DFA



18. Convert the following NFA to DFA.

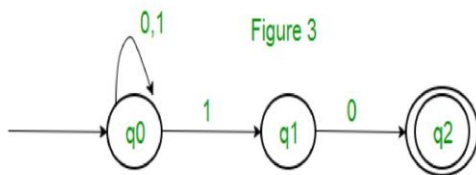
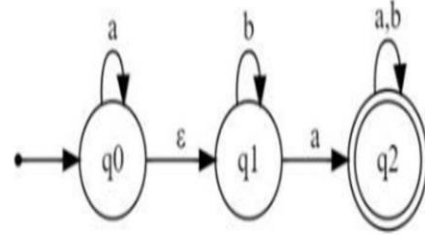


Figure 3

19. Convert Epsilon NFA to DFA



20. Convert the following NFA with ϵ to equivalent DFA.

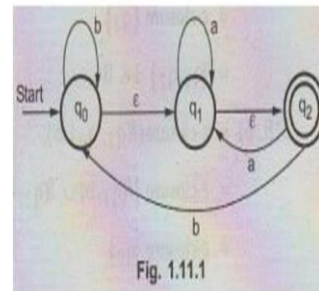


Fig. 1.11.1

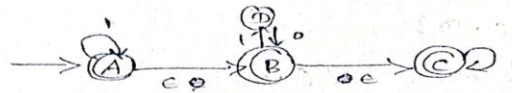
21. Consider following NFA with ϵ .

Input \ State	a	b	c	ϵ
p	$\{p\}$	$\{q\}$	$\{r\}$	$\varnothing$
q	$\{q\}$	$\{r\}$	$\varnothing$	$\{p\}$
r	$\{r\}$	$\varnothing$	$\{p\}$	$\{q\}$

Figure 3 Problems given to Students

14) Convert the following ϵ -NFA to NFA and then convert the resultant NFA to DFA.

POORNIMA A
JEEVA SRI M
PAVITHRA S



Soln:-

$$\epsilon\text{-closure}(\emptyset \cup A) = \{A, B, C\}$$

$$\epsilon\text{-closure}(B) = \{B, C\}$$

$$\epsilon\text{-closure}(C) = \{C\}$$

$$\epsilon\text{-closure}(\emptyset) = \{\emptyset\}$$

$$\begin{aligned} \delta'(A, 0) &= \hat{\delta}(A, 0) \\ &= \epsilon\text{-closure}(\delta(A, 0)) \\ &= \epsilon\text{-closure}(\delta(A, B, C), 0) \\ &= \epsilon\text{-closure}(\emptyset \cup \emptyset \cup \emptyset) \\ &= \emptyset \end{aligned}$$

$$\begin{aligned} \delta'(A, 1) &= \epsilon\text{-closure}(\delta(A, B, C), 1) \\ &= \epsilon\text{-closure}(A \cup \emptyset \cup C) \\ &= \{A, B, C\} \cup \{\emptyset\} \cup \{C\} \\ &= \{A, B, C, \emptyset\} \end{aligned}$$

$$\begin{aligned} \delta'(B, 0) &= \epsilon\text{-closure}(\delta(B, C), 0) \\ &= \epsilon\text{-closure}(\emptyset \cup \emptyset) = \emptyset \end{aligned}$$

$$\begin{aligned} \delta'(B, 1) &= \epsilon\text{-closure}(\delta(B, C), 1) \\ &= \epsilon\text{-closure}(\emptyset \cup \emptyset) \\ &= \emptyset \end{aligned}$$

$$\begin{aligned} \delta'(C, 0) &= \epsilon\text{-closure}(\delta(C, 0)) \\ &= \{\emptyset\} \end{aligned}$$

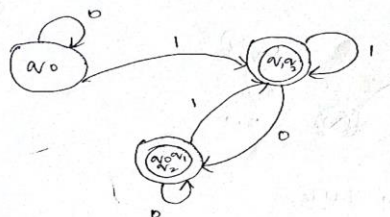
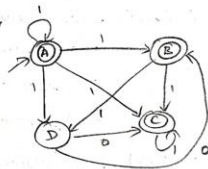
$$\begin{aligned} \delta'(C, 1) &= \epsilon\text{-closure}(\delta(C, 1)) \\ &= \{C\} \end{aligned}$$

$$\begin{aligned} \delta'(\emptyset, 0) &= \epsilon\text{-closure}(\delta(\emptyset, 0)) \\ &= \epsilon\text{-closure}(\emptyset) \\ &= \{\emptyset\} \end{aligned}$$

$$\begin{aligned} \delta'(\emptyset, 1) &= \epsilon\text{-closure}(\delta(\emptyset, 1)) \\ &= \{\emptyset\} \end{aligned}$$

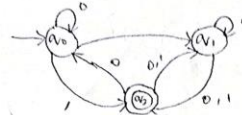
TT:-

State	Input	
	0	1
A	$\emptyset$	$\{A, B, C, \emptyset\}$
B	$\emptyset$	$\{C, \emptyset\}$
C	$\emptyset$	$\{C\}$
$\emptyset$	$\{B, C\}$	$\emptyset$



Convert the following NFA to DFA.

R. Priyadarshini
M. Sarawathi
F. Benitha Fency



Step-1:-

Transition table:-

A	0	1
q_0	q_0	q_1, q_2
q_1	q_1, q_2	q_2
q_2	q_0, q_1	q_1

Step 2:- DFA transition

A	0	1
q_0	q_0	q_1, q_2
q_1, q_2	q_0, q_1, q_2	q_1, q_2
q_0, q_1, q_2	q_0, q_1, q_2	q_1, q_2

Figure 4 Solution given by Students



Figure 5 Presentation by Divyasri J S and Parasuram C A

Reflective Critique:

❖ *Feedback of practice from students and other stakeholders:*

- Students expressed that they understood the concepts behind ϵ -NFA to NFA, ϵ -NFA to DFA and NFA to DFA.
- They found the exercise helpful in practicing the conversion of ϵ -NFA to NFA, ϵ -NFA to DFA and NFA to DFA.
- During the activity, they exchanged ideas on how to transform an ϵ -NFA to NFA, ϵ -NFA to DFA and NFA to DFA.
- Additionally, the exercise enhanced their presentation skills, oral communication, self-management, and leadership skills.

❖ *Benefit of the practice:*

- Students are able to recall the methods for converting an ϵ -NFA to NFA, ϵ -NFA to DFA and NFA to DFA.
- They can effectively share their ideas with their peers.
- The activity also supports slow learners in understanding the concepts better.

❖ *Challenges faced in implementation:*

- The activity was initially planned for 10 minutes but took 15 minutes to complete.
- Some students were hesitant to present their solutions.

References:

- ❖ <https://teaching.cornell.edu/teaching-resources/active-collaborative-learning/collaborative-learning>
- ❖ <https://www.nureva.com/blog/education/15-active-learning-activities-to-energize-your-next-college-class>

Signature of Faculty Member

HOD